The multinomial probability distribution and binomial distribution

CHAPTER 2

The multinomial probability distribution

The Multinomial Probability distribution

A binomial random variable results from an experiment consisting of n trials with two possible outcomes per trial. Frequently we encounter similar situations in which the number of possible outcomes per trial is more than two.

Ex:-

- Experiments that involve blood typing typically have at least four possible outcomes per trial.
- Experiments that involve sampling for defectives may categorize the type of defects observed into more than two classes.

A multinomial experiment is a generalization of the binomial experiment.

Definition 1

A multinomial experiment possesses the following **properties**:

1. The experiment consists of **n** identical trials.
2. The outcome of each trial falls into one of **k classes or cells**.
3. The probability that the outcome of a single trial falls into cell i, is p_i , **i = 1, 2 ,..., k** and remains the same from trial to trial. Notice that $p_1 + p_2 + p_3 + \cdots + p_k = 1$.
4. The trials are independent.
5. The random variables of interest are $Y_1, Y_2, \ldots, Y_k$, where Y_i equals the number of trials for which the outcome falls into cell i. Notice that $Y_1 + Y_2 + Y_3 + \cdots + Y_k = n$.

Definition 2

Assume that $\mathbf{p_1 + p_2 + p_3 + \cdots + p_k}$ are such that $\sum_{i=1}^{k} p_i = 1$, and $p_i > 0$ for $\mathbf{i = 1, 2, \ldots, k}$.

The random variables $\mathbf{Y_1, Y_2, \ldots, Y_k}$, are said to have a multinomial distribution with parameters n and $\mathbf{p_1 + p_2 + \cdots + p_k}$ if the joint probability function of $\mathbf{Y_1, Y_2, \ldots, Y_k}$ is given by

$$p(y_1, y_2, \ldots, y_k) = \frac{n!}{y_1! y_2! \cdots y_k!} p_1^{y_1} p_2^{y_2} \cdots p_k^{y_k},$$

where, for each i, $y_i = 0, 1, 2, \ldots, n$ and $\sum_{i=1}^{k} y_i = n$.

Extra Reading

Many experiments involving classification are multinomial experiments.
For example, classifying people into five income brackets results in an enumeration or count corresponding to each of five income classes. Or we might be interested in studying the reaction of mice to a particular stimulus in a psychological experiment. If the mice can react in one of three ways when the stimulus is applied, the experiment yields the number of mice falling into each reaction class. Similarly, a traffic study might require a count and classification of the types of motor vehicles using a section of highway. An industrial process might manufacture items that fall into one of three quality classes: acceptable, seconds, and rejects. A student of the arts might classify paintings into one of k categories according to style and period, or we might wish to classify philosophical ideas of authors in a study of literature. The result of an advertising campaign might yield count data indicating a classification of consumer reactions.
Many observations in the physical sciences are not amenable to measurement on a continuous scale and hence result in enumerative data that correspond to the numbers of observations falling into various classes.

Example 1

According to recent census figures, the proportions of adults (persons over 18 years of age) in the United States associated with five age categories are as given in the following table.

AGE	Proportion
18-24	0.18
25-34	0.23
35-44	0.16
45-64	0.27
Over 65	0.16

If these figures are accurate and five adults are randomly sampled, find the probability that the sample contains one person between the ages of 18 and 24, two between the ages of 25 and 34, and two between the ages of 45 and 64.

Theorem 1

If $\mathbf{Y_1}$, $\mathbf{Y_2}$,. . . , $\mathbf{Y_k}$ have a multinomial distribution with parameters n and $\mathbf{p_1} + \mathbf{p_2} + \mathbf{p_3} + \cdots + \mathbf{p_k}$, then

- $E(Yi) = np_i, \quad V(Yi) = np_iq_i$.
- $Cov(Ys, Y_t) = -np_sp_t \text{ , if } s \neq t.$

Q1

A learning experiment requires a rat to run a maze (a network of pathways) until it locates one of three possible exits. Exit 1 presents a reward of food, but exits 2 and 3 do not. (If the rat eventually selects exit 1 almost every time, learning may have taken place.) Let Yi denote the number of times exit i is chosen in successive runnings. For the following, assume that the rat chooses an exit at random on each run.

a) Find the probability that $n = 6$ runs result in $Y_1 = 3$, $Y_2 = 1$, and $Y_3 = 2$.

b) For general n, find $E(Y_1)$ and $V(Y_1)$.

c) Find $Cov(Y_2\ ,\ Y_3)$ for general n.

d) To check for the rat's preference between exits 2 and 3, we may look at $Y_2 - Y_3$. Find $E(Y_2 - Y_3)$ and $V(Y_2 - Y_3)$ for general n.

The bivariate normal distribution

Definition

- The multivariate normal density function is defined for k continuous random variables, Y_1, Y_2 ,..., Y_k . Because of its complexity, we will present only the bivariate density function (k = 2):

$$f(y_1, y_2) = \frac{e^{-Q/2}}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}, \qquad -\infty < y_1 < \infty,\ -\infty < y_2 < \infty,$$

where

$$Q = \frac{1}{1-\rho^2}\left[\frac{(y_1-\mu_1)^2}{\sigma_1^2} - 2\rho\frac{(y_1-\mu_1)(y_2-\mu_2)}{\sigma_1\sigma_2} + \frac{(y_2-\mu_2)^2}{\sigma_2^2}\right].$$

- If Y_1 and Y_2 have a bivariate normal distribution, they are independent if and only if their covariance is zero.